CSC 480 · MID-PROJECT HANDOUT · APRIL 13, 2026

Sentiment Analysis of Movie Reviews

Classical ML on the IMDb 50k dataset · Ethan Olla

91.12%
best test accuracy
Linear SVM + TF-IDF (uni+bi)

THE QUESTION

Can well-tuned classical ML classifiers reliably read a movie review's tone? The goal is to see how far plain scikit-learn pipelines get on IMDb binary sentiment before reaching for word embeddings or neural models.

SETUP

Dataset	IMDb 50k (Maas et al., 2011)
Split	25k train / 25k test, stratified
Classifiers	Naive Bayes, Logistic Reg., Linear SVM
Features	BoW, TF-IDF, TF-IDF + bigrams
Tuning	5-fold CV on training set
Cleaning	strip , lowercase

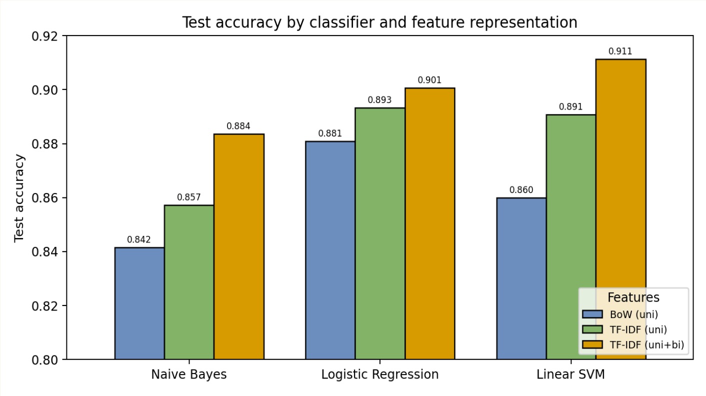
RESULTS

Model	BoW	TF-IDF	+Bigr.
Naive Bayes	0.842	0.857	0.884
Log. Reg.	0.881	0.893	0.901
Linear SVM	0.860	0.891	**0.911**

Test accuracy on the held-out 25k split. Best combination highlighted. Cross-validation agrees with test accuracy to within 0.005.

KEY FINDINGS

- **Bigrams help every model.** Naive Bayes jumps +2.6 points, suggesting that pairs like *not good* and *the worst* capture the negation patterns bag-of-words cannot see.
- **Linear SVM beats the 2011 baseline.** 91.12% vs. Maas et al.'s 88.9% on the same 25k test set, with no learned word vectors and no deep model.
- **LR and SVM are close.** Within one point across every feature setting. Ablations will separate them in the final report.



Test accuracy across three classifiers and three feature representations.

INSIDE THE MODEL

Top positive features from Logistic Regression: **great, excellent, perfect, amazing, hilarious, brilliant, wonderful, the best, loved, enjoyed**.

Top negative features: **bad, worst, awful, the worst, boring, poor, waste, terrible, stupid, horrible**.

The model's signal is exactly where you would expect it, and bigrams appear on both sides.

ERROR ANALYSIS

2,219 errors out of 25,000. Misclassified reviews average **235 words** vs. **229** for correct ones. Longer reviews more often mix tones and contain sarcasm, which is the working hypothesis for the remaining error budget.

WHAT'S NEXT

- **Error taxonomy** broken out by review length and negation presence.
- **Ablations** over vocabulary size, stop words, sublinear TF, and C.
- **Paired statistical tests** to separate Logistic Regression from SVM.
- **Writeup** in the NeurIPS LaTeX template, due May 10.

REFERENCES

- Maas et al., *Learning Word Vectors for Sentiment Analysis*, ACL 2011.
- Pang, Lee, Vaithyanathan, *Thumbs up?*, EMNLP 2002.
- Wang and Manning, *Baselines and Bigrams*, ACL 2012.